OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION: SALEM

SYSTEM: Emergency Procedures

TASK: Take Corrective Action For A Nuclear Instrumentation System Malfunction (Energize Source Range NIS IAW TRIP-2 w/one under compensated IR channel)

TASK NUMBER: N1140230401

JPM NUMBER: 14-01 NRC Sim f

ALTERNATE PATH: [X]

K/A NUMBER: EPE 007 EA1.05

IMPORTANCE FACTOR: 4.0 (RO) 4.1 (SRO)

APPLICABILITY:
EO [ ] **RO** [X] **STA** [ ] **SRO** [X]

EVALUATION SETTING/METHOD: Simulator / Perform

REFERENCES: 2-EOP-TRIP-2, Rev. 28 Reactor Trip Response (rev checked 9-23-15)

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 5 Minutes

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By: **G Gauding** **Instructor** **Date: 8-22-15**

Validated By: **S Bickhart / J Militti** **SME or Instructor** **Date: 9-23-15**

Approved By: [signature] **Training Department** **Date:** 10-7-15

Approved By: [signature] (mob) **Operations Department** **Date:** 10 9 15

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY:
GRADE: [ ] **SAT** [ ] **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

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TASK: Take Corrective Action For A Nuclear Instrumentation System Malfunction (Energize Source Range NIS IAW TRIP-2 One under compensated IR channel)

TASK NUMBER: 1150030501

SIMULATOR SETUP: **IC-246**

MALF: NI0195D IR CH N36 Compensating Volts Lo - TRUE
Mark up EOP-2 up to Step 19 of TRIP-2
Ensure NR-45 set up for SR1 on PEN 1 and IR2 on PEN 2
Ensure audio count rate monitor selected to 10K scale

INITIAL CONDITIONS:

A MANUAL reactor trip was initiated 15 minutes ago when both SGFP's tripped automatically. Operators performed the immediate actions of EOP-TRIP-1, Reactor Trip or Safety Injection, then transitioned to 2-EOP-TRIP-2, Reactor Trip Response. Operators have performed TRIP-2 up to Step 19, Steam Dump Mode Shift.

INITIATING CUE:

You are the board operator. Starting at Step 19, perform 2-EOP-TRIP-2 Reactor Trip Response

Successful Completion Criteria:
1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Performance for Successful Completion:
1. Transfers Main Steam Dumps to MS Pressure Control – Auto
2. Energizes BOTH SRNI Channels

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Note for Evaluator: ~ 2 minutes 10 seconds will elapse between starting JPM and when IR CH II amps will level off.		
	19	Are Condenser Steam Dumps Available?	Checks Condenser Steam Dumps and reports they are available based on circulators in service and condenser vacuum established.		
*		Place Steam Dumps in "Manual"	Depresses Steam Dumps Manual PB on control console and verifies Manual light illuminates and Auto light extinguishes.		
*		Align Steam Dump Valve demand "Press %" and "Tavg %"	Uses Increase Demand (Open Vlv) PB on to align Steam Dump Valve demand "Press %" and "Tavg %".		
*		Place Steam Dumps in "MS Pressure Control"	Depresses "MS Pressure Control" PB on control console and verifies light illuminates and Tavg Control light extinguishes.		
*		Place Steam Dumps in "Auto"	Depresses Steam Dumps Auto PB on control console and verifies Auto light illuminates and Manual light extinguishes.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Adjust Steam Pressure Valve Demand to maintain SG Pressure at 1005 psig.	**Cue:** CRS directs you to leave Steam Dumps at current pressure setpoint.		
	20	Is <u>any</u> RCP running	Checks RCP status and determines all RCP's are running.		
	22	Are <u>both</u> IR Channels less than 7E-11 Amps	Checks IRNI Channel 1 and Channel II indication and determines 2N36 reads >7E-11 Amps.		
		Is undercompensation preventing proper IR operation?	Determines undercompensation of channel 2N36 is preventing proper IR operation by: - Elapsed time since trip - SUR 0 on affected channel with power above minimum display - NR-45 trend IR CH II leveling off **<u>If</u> operator arrives quickly at this step and continues without determining overcompensation problem, <u>then</u> when step 24 is reached (below)about whether a C/D will be perfomed give operator cue (below) that SM is in contact with Ops Manager about unit status and CRS is waiting for direction.**		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
 * *	22.1	Energize Source Range Channels	Energizes Source Range Channel I by depressing RESET SOURC RANGE A Energizes Source Range Channel II by depressing RESET SOURC RANGE B		
	22.2	Transfer NR-45 (Nuclear Power Recorder) to Source Range Channels	Transfers NR-45 (Nuclear Power Recorder) to Source Range Channels by selecting Source Range Channel I on Pen 1 or 2, and selecting Source Range Channel II on the other Pen.		
	22.3	Adjust Audio Count Rate Circuit Scale	Adjusts Audio Count Rate Circuit Scale		
			Terminate JPM after audio count rate scale adjustment has been verified in the control room.		
	Step 23 contingency	Maintain plant parameters as follows: PZR Pressure – 2235 psig PZR level -22% SG Level – 9%-33% RCS Temp - 547°F	Ensures plant parameters are maintained.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	Step 24 contingency	Is RCS Cooldown Required	**Cue**: SM is in contact with Ops Manager about unit status and CRS is holding at this step waiting for direction.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

GS 1. Task description and number, JPM description and number are identified.

GS 2. Knowledge and Abilities (K/A) references are included.

GS 3. Performance location specified. (in-plant, control room, or simulator)

GS 4. Initial setup conditions are identified.

GS 5. Initiating and terminating Cues are properly identified.

GS 6. Task standards identified and verified by SME review.

GS 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

GS 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 28 Date 8/25/11

GS 9. Pilot test the JPM:
   a. verify Cues both verbal and visual are free of conflict, and
   b. ensure performance time is accurate.

________ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

________ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [signature] BURKHART Date: 9/23/15

SME/Instructor: [signature] M.Leith Date: 9-23-15

SME/Instructor: ________________ Date: ________

INITIAL CONDITIONS:

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